

# Vali Hameed

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## Education

### Lancaster University

September 2024 – June 2027

- Computer Science BSc with Honours (Year 2) - Expected First Class
- First Year Average - 70%

## Experience

### Co-Founder & Lead Engineer, Picky Eater – Lancaster, ENG

October 2025 – June 2026

- Co-founded and led a 7-person team as the **top contributor (60% of commits)**, architecting a full-stack platform across web (**React/Vite**), mobile (**React Native**), and backend (**Django REST**).
- Designed a **polyglot data architecture** using **MongoDB** for social content and **PostgreSQL** for relational data, balancing NoSQL flexibility with strict **ACID guarantees**.
- Built a **7-signal recommendation algorithm** using order history, recency decay, dietary matching, and MongoDB aggregations, achieving **O(n log k)** selection performance.
- Engineered an **API security framework** (JWT auth, rate limiting, input sanitization) and authored regression test suites to harden against **OWASP Top 10** vulnerabilities.
- Implemented **real-time WebSocket chat** via **Django Channels/Redis**, and established PR workflows and CI/CD pipelines to increase milestone completion by **15%**.

### Software Engineering Intern, DigbySwift – Leeds, ENG

June 2023

- Developed a **cross-platform mobile application** from concept to completion using **Dart** and **Flutter**, delivering a user-centric movie and TV series search tool and **reduced** app search time by **25%**.
- Ensured code reliability and software quality by implementing comprehensive **unit tests**.
- Managed project version control using **Git** and **GitHub**, employing a feature-branching workflow with pull requests for code review before merging into the main branch.

## Projects

### FightPicks

- **Architected a microservices platform** across three repositories, integrating a **Next.js 15/TypeScript** frontend with a **Java Spring Boot REST API** and two **Python FastAPI** services.
- **Trained and deployed a machine learning model** (Gradient Boosting Classifier) using scikit-learn, achieving **66.09%** true symmetric accuracy across **7,000+** historical fights, deploying the containerized inference service to **AWS ECS**.
- **Engineered secure, scalable authentication** via **Spring Security**, implementing stateless **JWTs** with refresh token rotation, token-bucket rate limiting, and **Flyway**-managed **PostgreSQL** migrations.
- **Developed an autonomous Python scraper** (BeautifulSoup/Playwright) to continuously feed live fighter and event data into the core API, powering real-time leaderboards.
- **Successfully launched the platform to live users** using a hybrid deployment strategy, hosting the core stack on an **Oracle VPS** while scaling the ML service on **AWS ECS**, managed via automated **GitHub Actions CI/CD pipelines**.

### Orbital Risk

- **Architected a mission control dashboard** using **Next.js 14** (a **React** framework) and **TypeScript**.
- **Engineered a real-time 3D visualisation** system to map satellite orbits and debris fields, using **WebGL** and **Three.js**. Enabling interactive trajectory analysis for safe launch planning.
- **Implemented a custom risk assessment engine** that calculates collision probabilities by processing orbital parameters (altitude, inclination, corridor width, time of launch) against real-time debris density datasets.
- **Developed a Python-based weather prediction service** to analyse atmospheric conditions and generate automated Go/No-Go launch recommendations through **FastAPI** endpoints and an **XGBoost** model.
- **Optimized data retrieval and model execution** by implementing a local **SQLite** caching layer within the backend architecture, reducing redundant data fetching and **improving dashboard load times**.

## Technical Skills

**Languages:** Java, Python, Dart, JavaScript, TypeScript, SQL, C, C++, Haskell

**Frameworks/Libraries:** Flutter, Next.js, Tailwind CSS, React, Pandas, Scikit-learn, FastAPI, Spring Boot, Django

**Tools & platforms:** Git/GitHub, Unit Testing, Docker, AWS, PostgreSQL, MongoDB, MySQL, Vercel, Render, Oracle